

IMPERIAL

Carbon Accounting and Reporting

Lecture 1 – Course Intro / Big Picture

Dr James Robey / Jon Hampson
12 September 2025

Course Overview

COURSE OVERVIEW

- I. Sustainability – the Big Picture
- II. Carbon and Climate Change
- III. Basics of Carbon Accounting
- IV. Introduction to Organisational Carbon Accounting
- V. Scope 2 – energy emissions
- VI. Scope 3 – indirect emissions
- VII. Net Zero, Science Based Targets and Carbon Credits
- VIII. Sustainability reporting & materiality

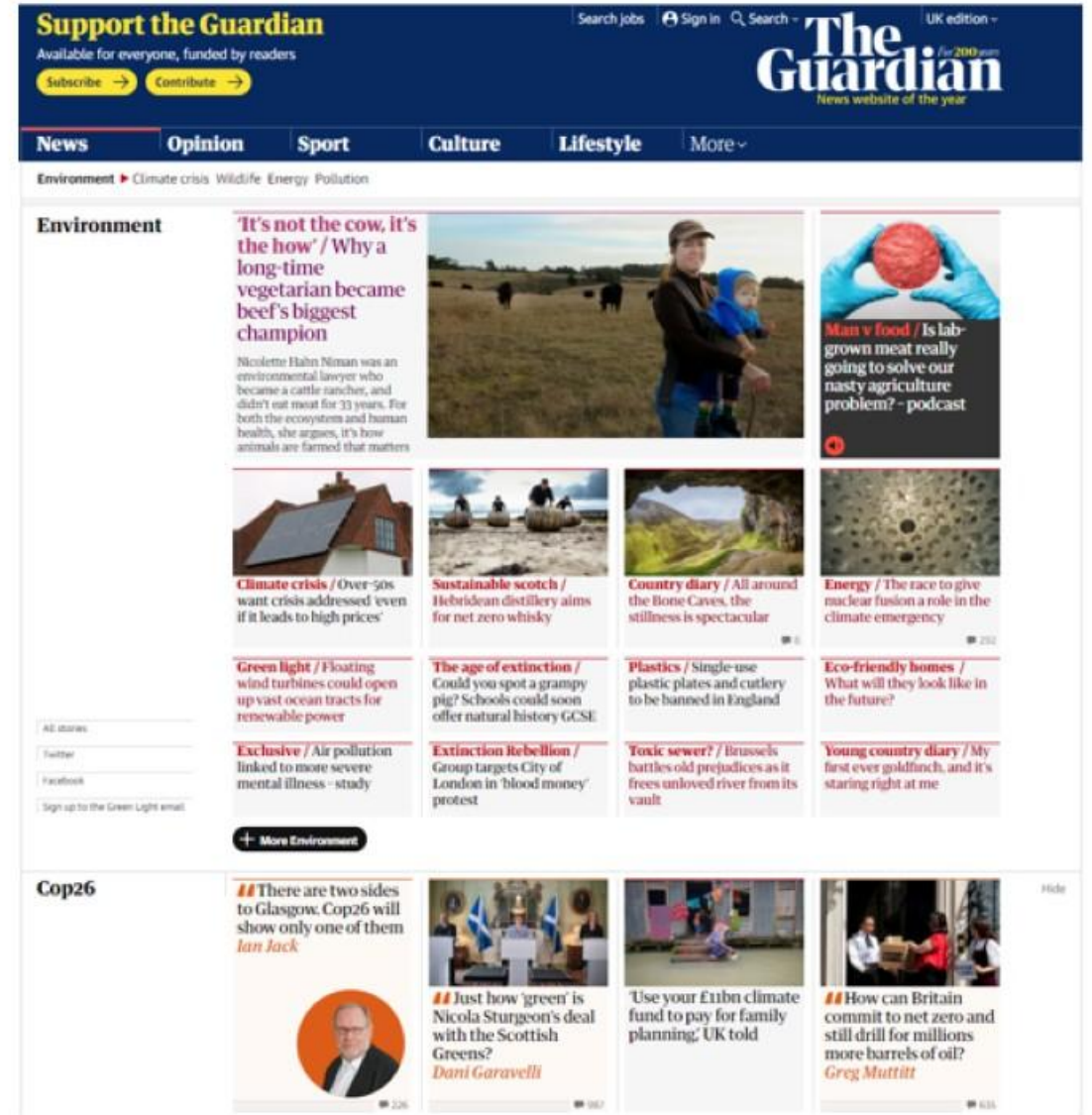
Useful reading and research sources:

Reading

- Financial Times
- The Guardian
- The Economist
- Forbes / Bloomberg / Reuters
- Edie / Business Green / greendigest.co

Research Sources

- The Economist (EIU)
- CIA data base
- World Bank, OECD, UN
- CDP



Reference list (1/2) – General Sustainability

Recommended Articles

- Arrow, K., Dasgupta, P., Goulder, L., Daily, G., Ehrlich, P., Heal, G., Levin, S., Mäler, K.-G., Schneider, S., Starrett, D. and B. Walker (2004) 'Are We Consuming Too Much?' *Journal of Economic Perspectives*, 18 (3), 147-172.
- Braungart, M., McDonough, W. and A. Bollinger (2007) 'Cradle-to-cradle design: creating healthy emissions - a strategy for eco-effective product and system design' *Journal of Cleaner Production*, 15 (13/14), 1337-1348.
- Carroll, A.B. and K.M. Shabana (2010) 'The Business Case for Corporate Social Responsibility: A Review of Concepts, Research and Practice' *International Journal of Management Reviews*, 12(1), 85-105.
- Hart, S. (1995) 'A Natural-Resource-Based View of the Firm' *Academy of Management Review*, 20 (4), 986-1014.
- Lovins, A.B., L.H. Lovins and P. Hawken (1999) 'A Road Map for Natural Capitalism' *Harvard Business Review*, May/June, 145-158.

Additional Reading

- Anderson, R. and R. White (2009) *Confessions of a Radical Industrialist*, London: Random House Business Book.
- Economist (various years) *Pocket World in Figures*, London: The Economist.
- Elkington, J. (1999) *Cannibals with Forks: The Triple Bottom Line of 21st Century Business*, Oxford: Capstone Publishing Ltd.
- Hart, S. (2005) *Capitalism at the Crossroads: The Unlimited Business Opportunities in Solving the World's Most Difficult Problems*, Upper Saddle River, New Jersey: Wharton School Publishing.
- Hawken, P., Lovins, A.B. and L.H. Lovins (2001) *Natural Capitalism: The Next Industrial Revolution*, London: Earthscan Publications Ltd.
- Hawken, Paul (ed.) (2017) *Drawdown: The Most Comprehensive Plan Ever Proposed to Reverse Global Warming*, London: Penguin Books.
- Klein, Naomi (2015) *This Changes Everything: Capitalism vs. the Climate*, London: Penguin.
- Lynas, Mark (2020) *Our Final Warning: Six Degrees of Climate Emergency*, London: 4th Estate.
- McDonough, W. and M. Braungart (2002) *Cradle to Cradle – Remaking the Way we Make Things*, New York: North Point Press.
- Porritt, J. (2007) *Capitalism as if the World Matters (Revised Edition)*, London: Earthscan.
- Porritt, Jonathon (2020) *Hope in Hell: A decade to confront the climate emergency*, Simon & Schuster, London.
- Raworth, Kate (2018) *Doughnut Economics: Seven Ways to Think Like a 21st Century Economist*, London: Random House Business.
- Smil, Vaclav (2019) *Growth: From Microorganisms to Megacities*, Cambridge Mass, MIT Press.
- Victor, P.A. (2008) *Managing Without Growth: Slower by Design, Not Disaster*, Cheltenham: Edward Elgar Publishing Limited.

Reference list (2/2) – Carbon Accounting

Useful References

CDP (2024) Various reports <https://www.cdp.net/en/research>

Enkvist, P. et al, "A cost curve for greenhouse gas reduction". McKinsey Quarterly, 2007 Number 1

Greenhouse Gas Protocol, "Product Life Cycle Accounting and Reporting Standard", World Resources Institute and World Business Council for Sustainable Development, Pankaj Bhatia, Cynthia Cummis, Laura Draucker, David Rich, Holly Lahd (WRI), Andrea Brown (WBCSD), October 2011 (pages 1-148) http://pdf.wri.org/ghgp_product_life_cycle_standard.pdf

IPCC (2006), 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Prepared by the National Greenhouse Gas Inventories Programme, Eggleston H.S., Buendia L., Miwa K., Ngara T. and Tanabe K. (eds)., Overview http://www.ipcc-nggip.iges.or.jp/public/2006gl/pdf/0_Overview/V0_1_Overview.pdf

IPCC (2014) "Climate Change 2014 Synthesis Report (AR5)" https://www.ipcc.ch/site/assets/uploads/2018/02/SYR_AR5_FINAL_full.pdf

IPCC (2014) "Summary for Policymakers" http://www.ipcc.ch/pdf/assessment-report/ar5/wg1/WG1AR5_SPM_FINAL.pdf

IPCC (2021) "Climate Change 2021 – The Physical Science Basis: Summary for Policymakers" https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC_AR6_WGI_SPM.pdf

IPCC (2023) "Climate Change 2023 Synthesis Report (AR6)" <https://www.ipcc.ch/report/ar6/syr/>

Project Drawdown (2024) <https://www.drawdown.org/>

Stern, Nicolas (2006) "Stern Review on the Economics of Climate Change" (Cambridge University Press: Cambridge, United Kingdom), October 20. (Executive Summary)

U.K. Government (2019) "Environmental reporting guidelines: including Streamlined Energy and Carbon Reporting requirements" <https://www.gov.uk/government/publications/environmental-reporting-guidelines-including-mandatory-greenhouse-gas-emissions-reporting-guidance>

U.S. EPA (2023) Greenhouse Gas Reporting Program: <https://www.epa.gov/ghgreporting>

World Resources Institute and World Business Council for Sustainable Development (2004) "The Greenhouse Gas Protocol: A Corporate Accounting and Reporting Standard (revised edition)" <https://ghgprotocol.org/sites/default/files/standards/ghg-protocol-revised.pdf>

World Resources Institute and World Business Council for Sustainable Development (2011) "Greenhouse Gas Protocol: Corporate Value Chain (Scope 3) Accounting and Reporting Standard", (pages 1-152) https://ghgprotocol.org/sites/default/files/standards/Corporate-Value-Chain-Accounting-Reporting-Standard_041613_2.pdf

Course Assessment

Course Assessment:

ASSESSMENT

Group Assignment 90%

- Full details provided in week 4/5
- 60% based on report
- 40% based on group presentation

Participation 10%

- Lecture participation / homework exercises
- Participation in the final presentation Q&A



Presentation dates will be:

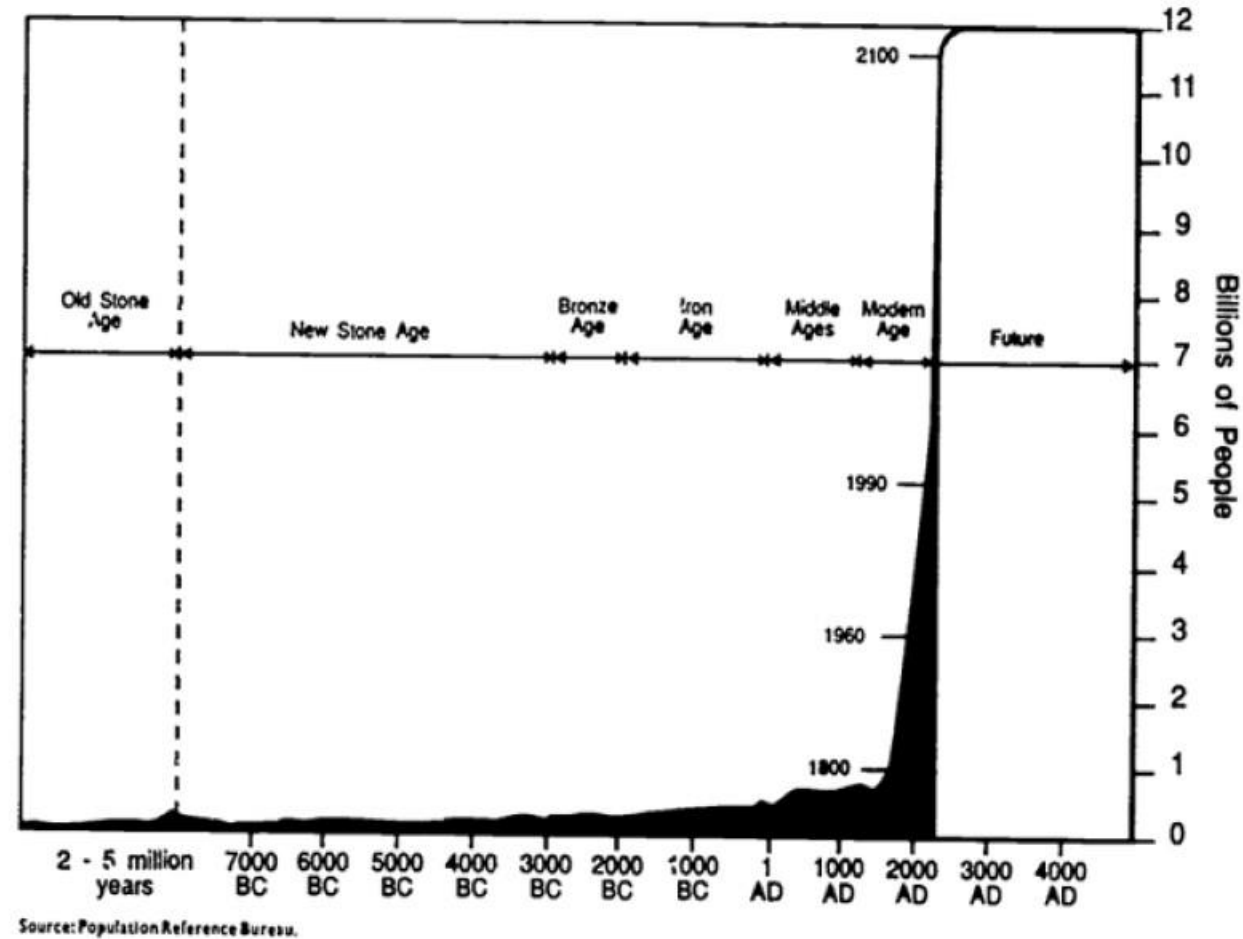
- Thurs 27th November
- Friday 28th November

Sustainability Context



Global population ...

Figure 9
World Population Growth through History



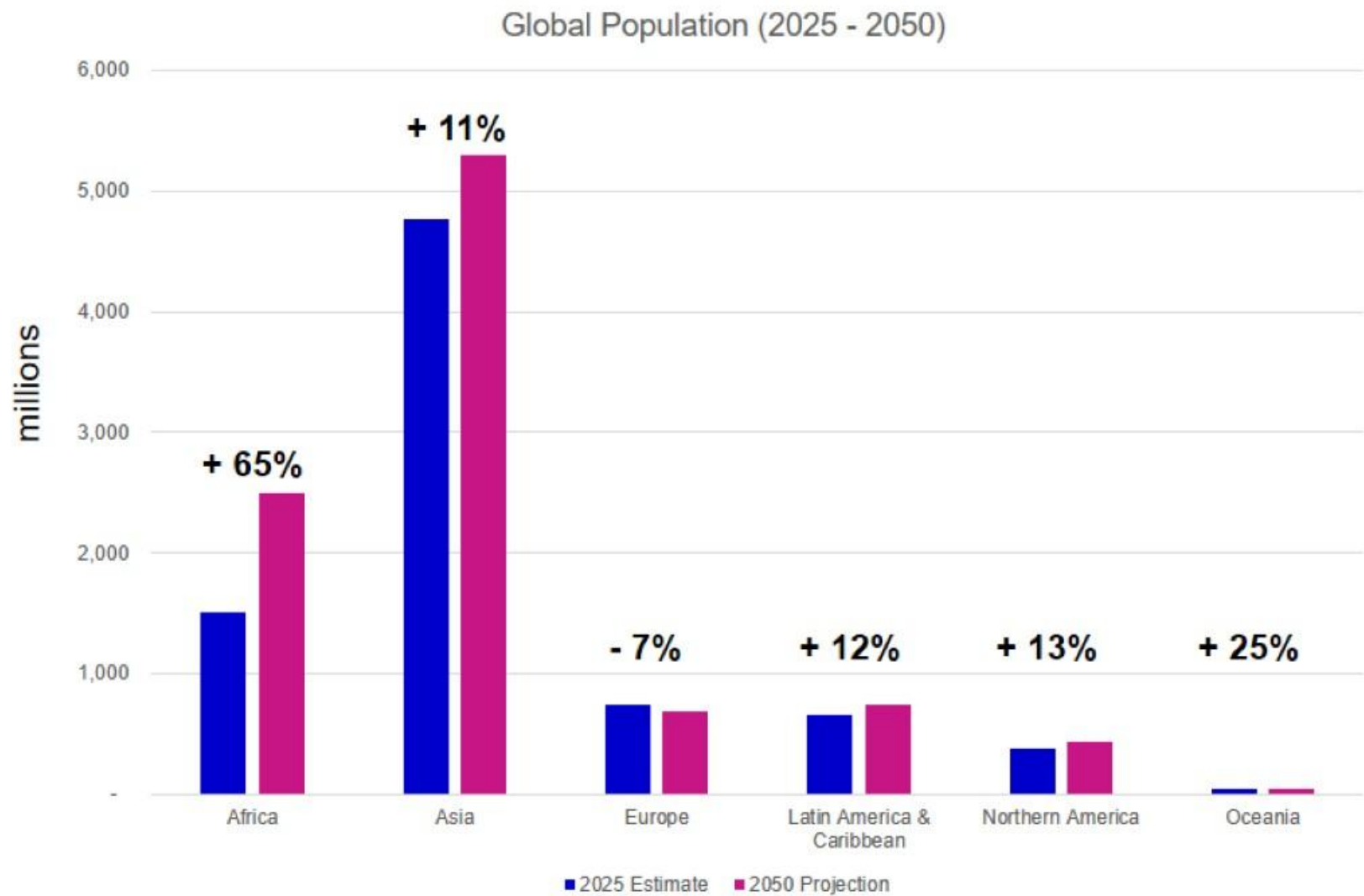
Source: Population: A Lively Introduction,* Joseph A. McFall, Jr., Population Bulletin, Volume 46, No. 2, 1991, pages 1-43, Population Reference Bureau, Washington DC

“...world population is expected to reach 9.7 billion in 2050 and could peak at nearly 11 billion around 2100”

UN Department of Economic and Social Affairs

<https://www.un.org/development/desa/en/news/population/world-population-prospects-2019.html>

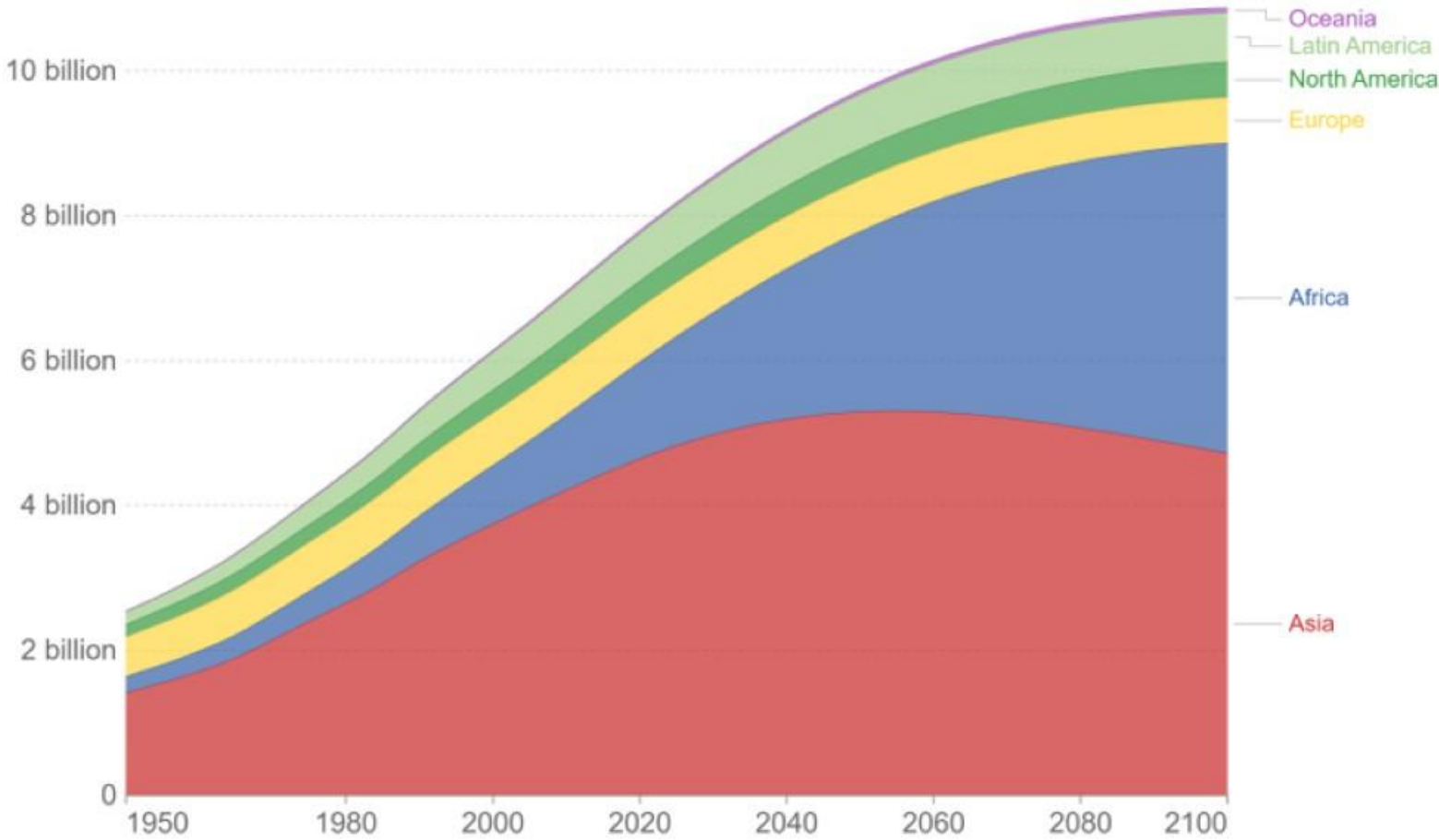
Global population ...



Global population ...

World population by region, 1950 to 2100

Projected population to 2100 is based on the UN's medium population scenario.



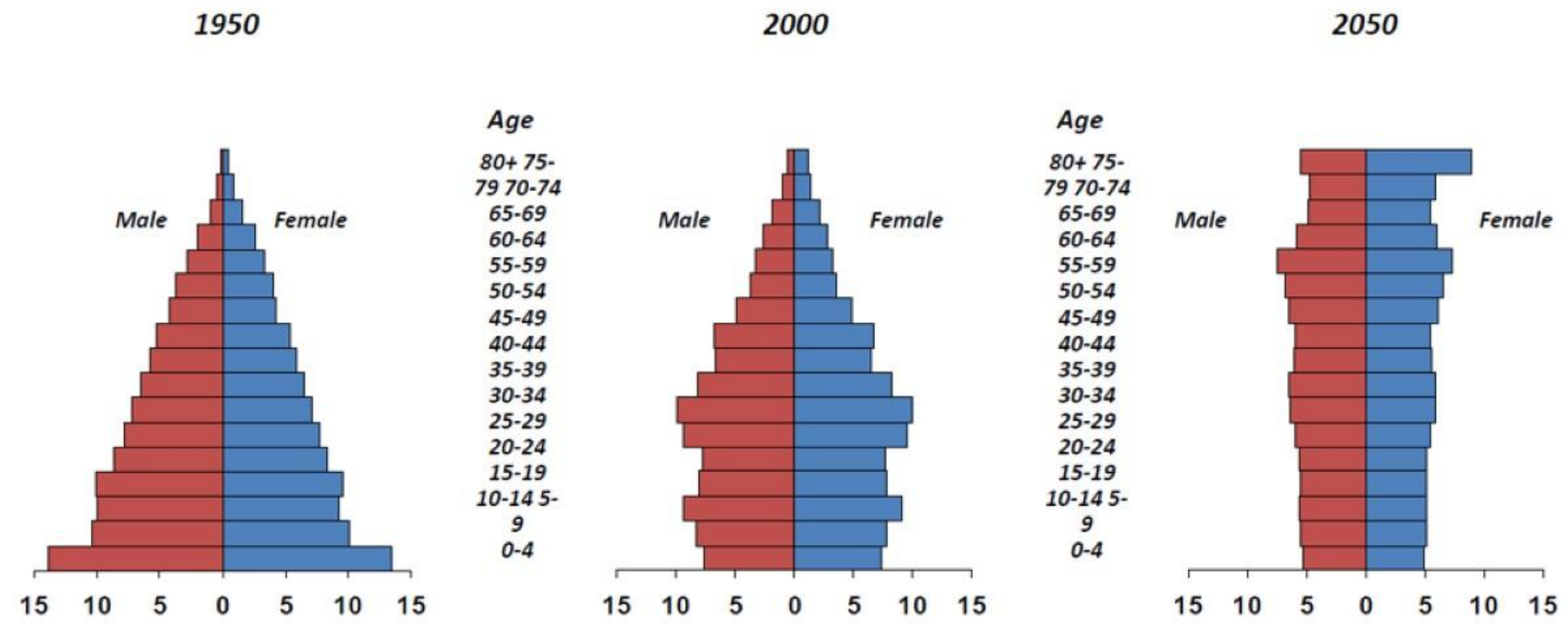
Source: HYDE (2016) & UN, WPP (2019)

OurWorldInData.org/world-population-growth • CC BY

<https://ourworldindata.org/grapher/historical-and-projected-population-by-region>

Shifting Demographics of China

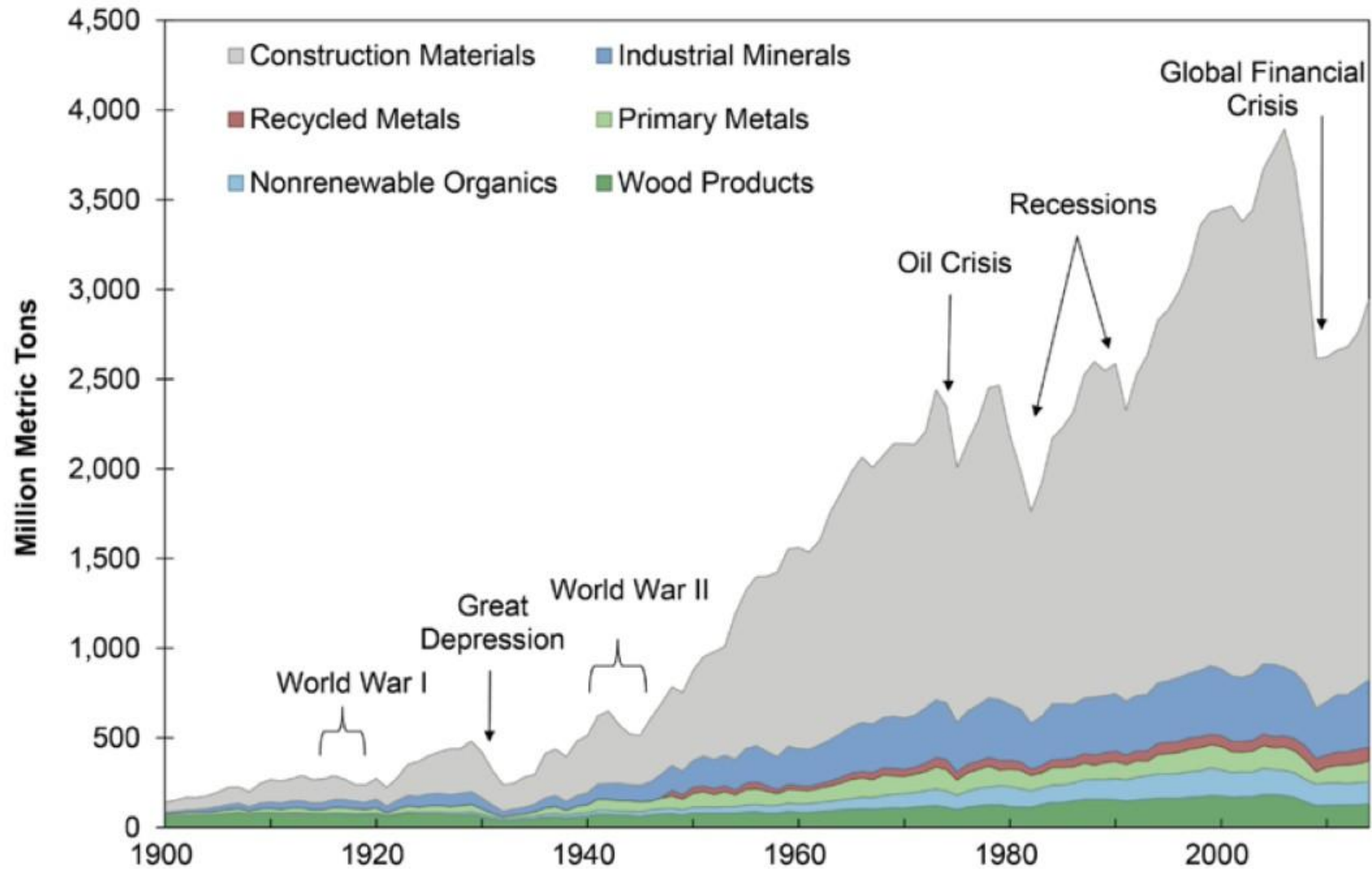
Population Structures by Age and Sex
Millions



Source: World Population Prospects: The 2004 Revision (2005).

Materials through-put in the Global Economy

U.S. Nonfuel Material Consumption, 1900-2014¹



<http://css.umich.edu/factsheets/us-material-use-factsheet>

Rare Earth Elements

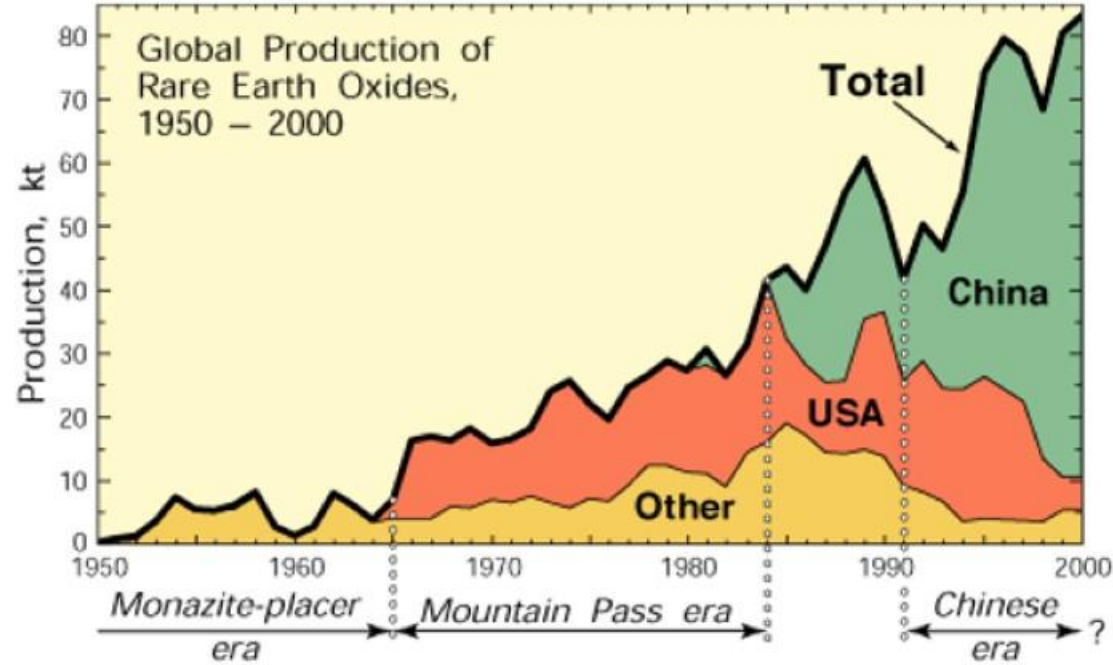


Table 1. Names and symbols of the REE

La	lanthanum	Tb	terbium
Ce	cerium	Dy	dysprosium
Pr	praseodymium	Ho	holmium
Nd	neodymium	Er	erbium
Pm	promethium	Tm	thulium
Sm	samarium	Yb	ytterbium
Eu	europium	Lu	lutetium
Gd	gadolinium	Y	yttrium

<http://pubs.usgs.gov/fs/2002/fs087-02/>

‘Consumption vs ‘circularity’

Only 6.9% of the materials entering the global economy are secondary.

Extraction of raw materials to rise by 60% by 2060, says UN report

Exclusive: Report proposes action to reduce overall demand rather than simply increasing ‘green’ production



📍 A mine near Kolwezi in the Democratic Republic of the Congo. Photograph: Junior Kannah/AFP/Getty Images

www.theguardian.com/environment/2024/jan/31/raw-materials-extraction-2060-un-report

The Circularity Gap Report 2024

"In the last five years, humanity consumed a whopping 500 billion tonnes of materials—*nearly equal to what was consumed during the entire 20th century.*

The global circularity rate has fallen steadily *from 9.1% in 2018*, when Circle Economy Foundation began measuring, *to 6.9% in 2024.*

This means that out of all the materials consumed worldwide, we’re consuming more virgin materials than ever—while the share of secondary materials is in decline.“

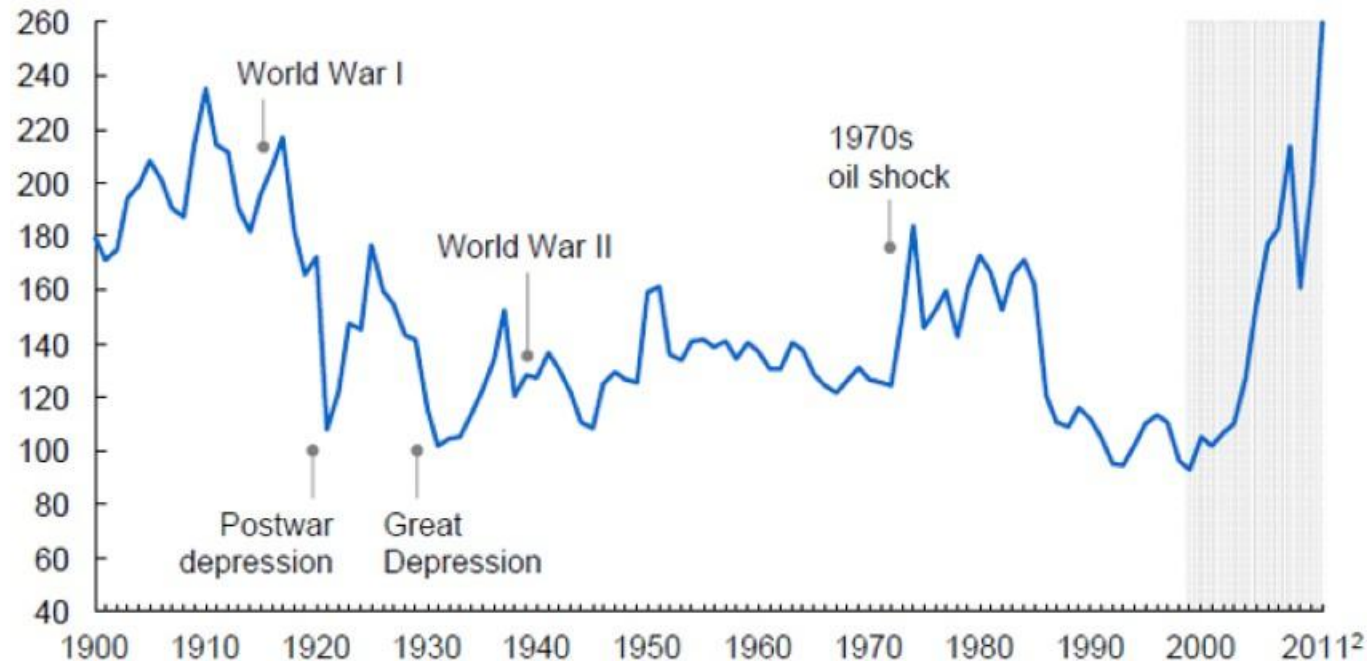
<https://global.circularity-gap.world/>

Commodity Prices

Exhibit E1

Commodity prices have increased sharply since 2000, erasing all the declines of the 20th century

MGI Commodity Price Index (years 1999–2001 = 100)¹



¹ See the methodology appendix for details of the MGI Commodity Price Index.

² 2011 prices are based on average of the first eight months of 2011.

SOURCE: Grilli and Yang; Stephan Pfaffenzeller; World Bank; International Monetary Fund (IMF); Organisation for Economic Co-operation and Development (OECD); UN Food and Agriculture Organization (FAO); UN Comtrade; McKinsey analysis

**Commodity Price increase
2000 – 2014 (MGI)**

Metals 259%

Lead	1184%
Gold	446%
Copper	390%
Platinum	297%
Nickel	156%

Energy 346%

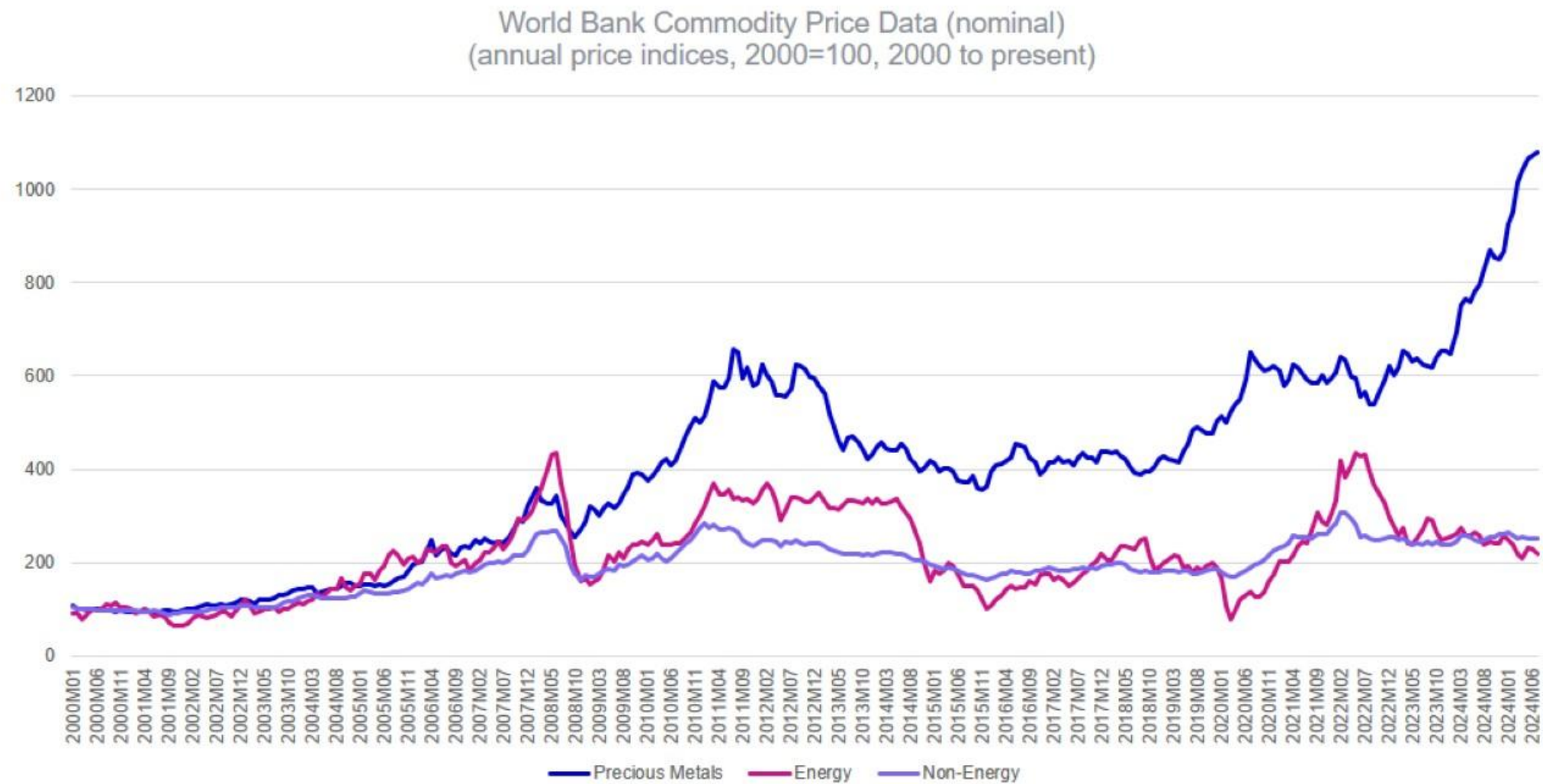
Oil	375%
Uranium	374%
Coal	317%
Gas	270%

Food 189%

Corn	222%
Bananas	190%
Coffee	167%
Sugar	145%

Source: McKinsey Global Institute (2011) 'Resource Revolution: Meeting the world's energy, materials, food, and water needs'

Commodity Prices

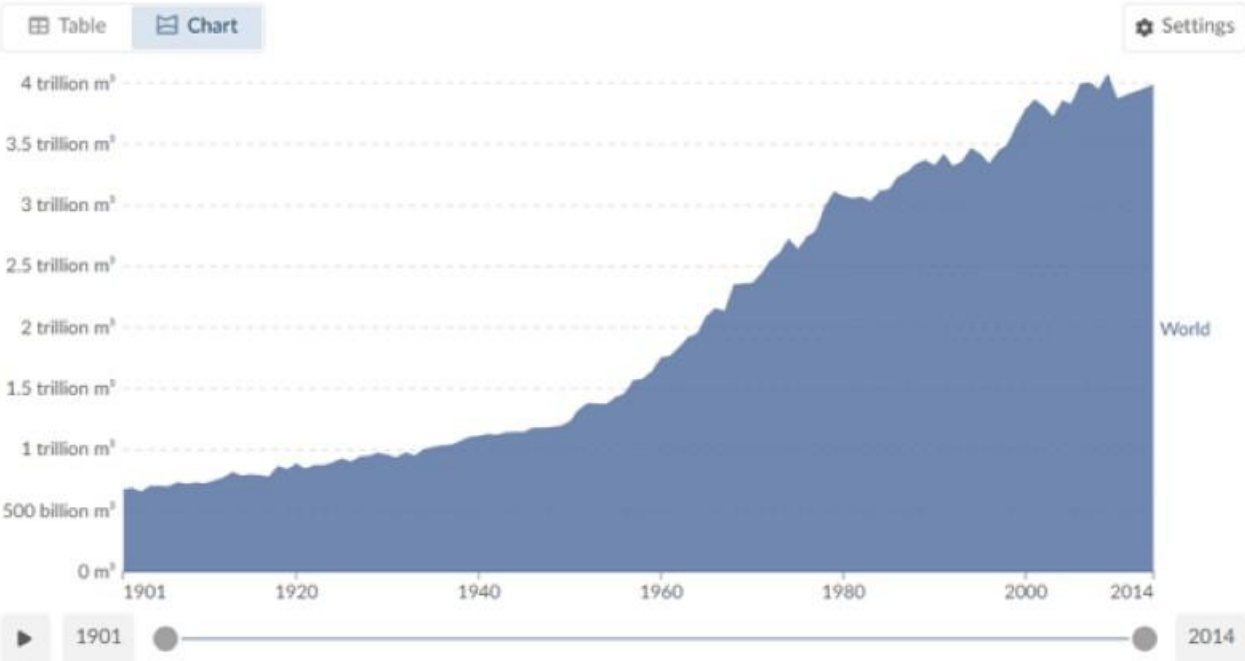


<https://www.worldbank.org/en/research/commodity-markets>

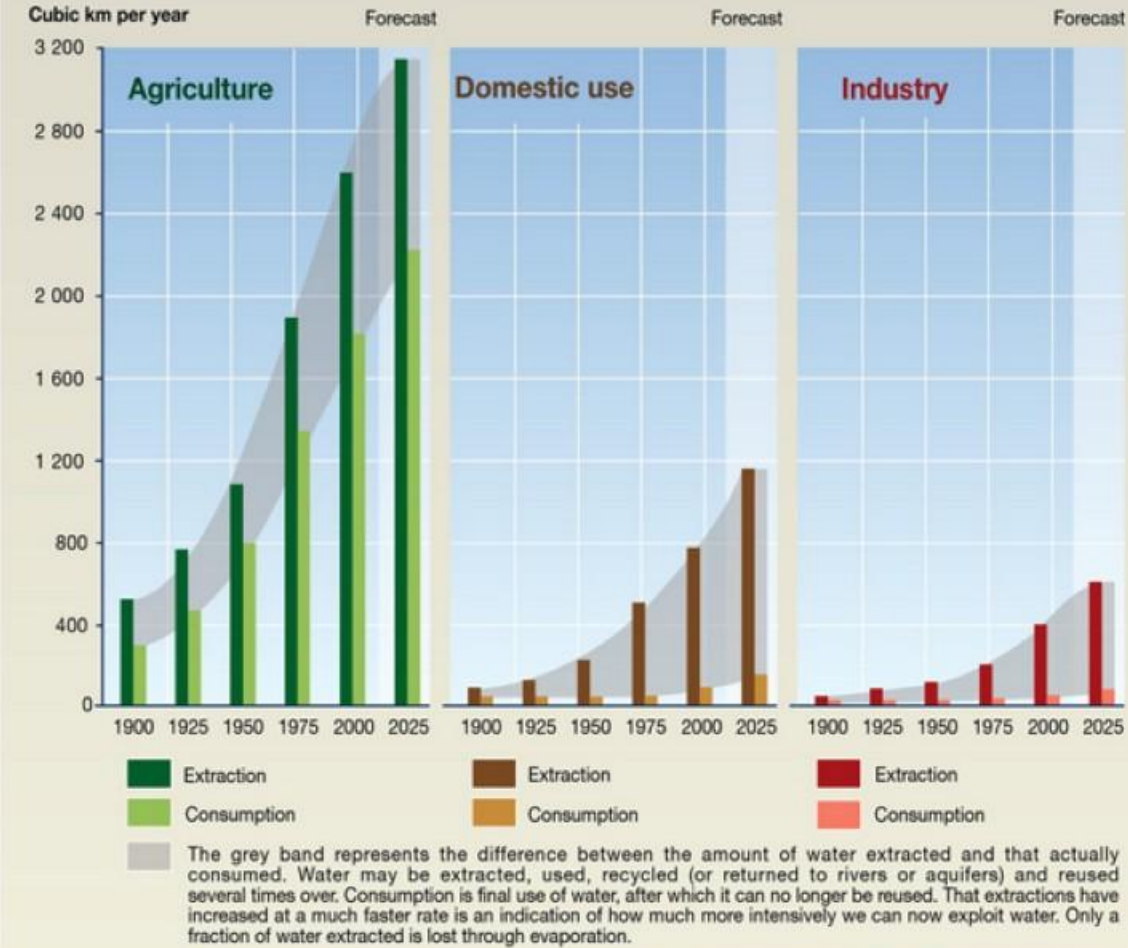
Projected water consumption by sector

Global freshwater use over the long-run

Global freshwater withdrawals for agriculture, industry and domestic uses since 1900, measured in cubic metres (m³) per year.



<https://ourworldindata.org/water-use-stress>
<http://www.unep.org/dewa/vitalwater/jpg/0211-withdrawcons-sector-EN.jpg>

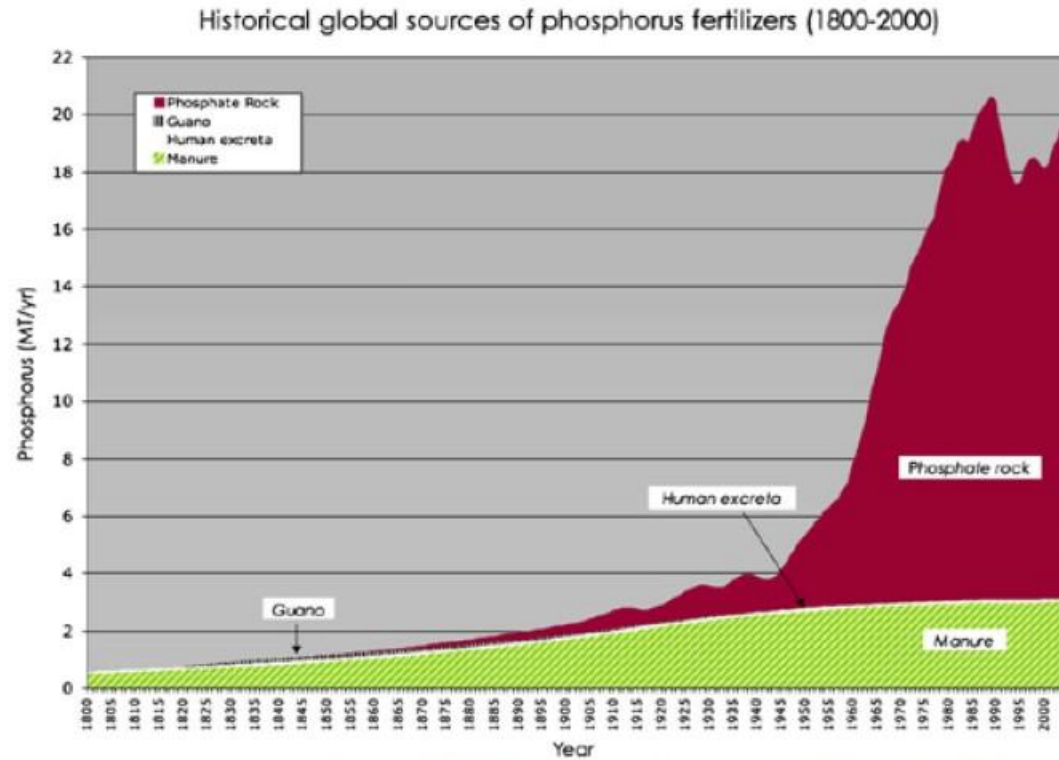


Source: Igor A. Shiklomanov, State Hydrological Institute (SHI, St. Petersburg) and United Nations Educational, Scientific and Cultural Organisation (UNESCO, Paris), 1999.

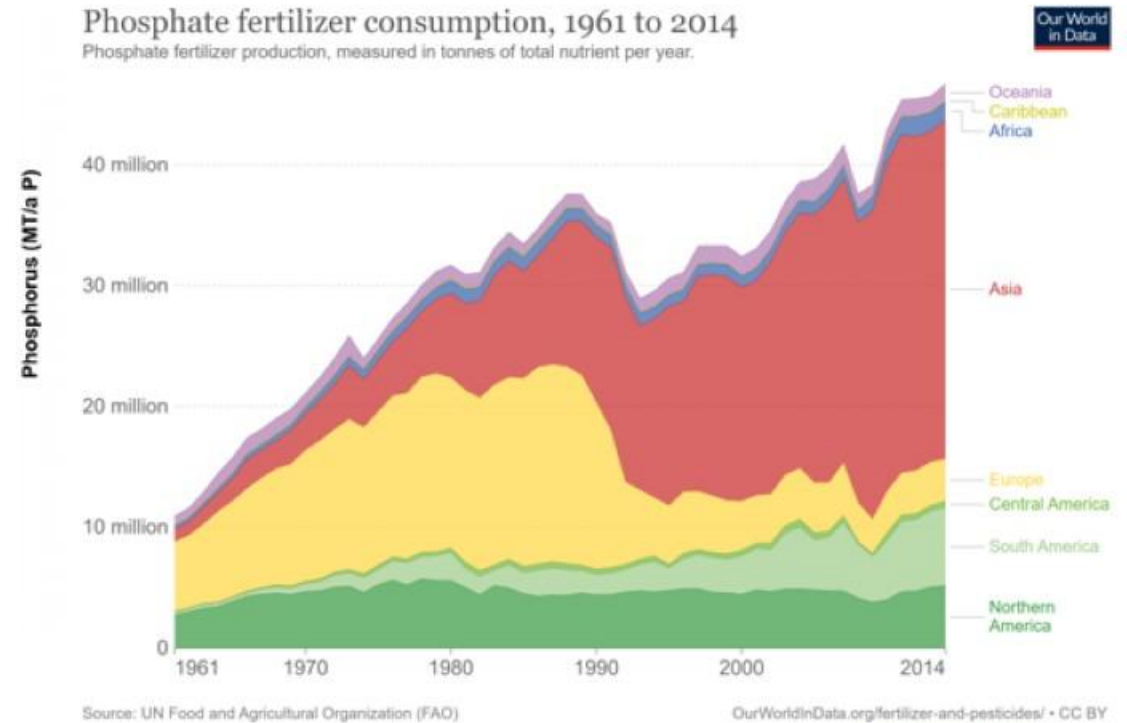




Use of fertilizers is an indication of intensive farming techniques



Source: Cordell et al (2009) 'The story of phosphorus: Global food security and food for thought' *Global Environmental Change* Volume 19, Issue 2, May 2009, pp.292-305



By the 2050s, one-third of the world's rivers could be affected by water scarcity or be polluted by nitrogen.



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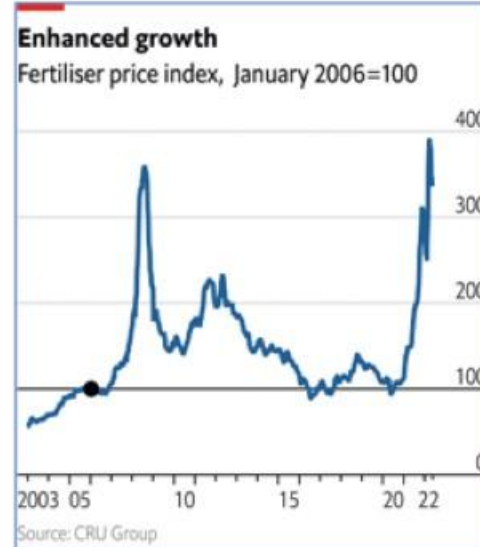
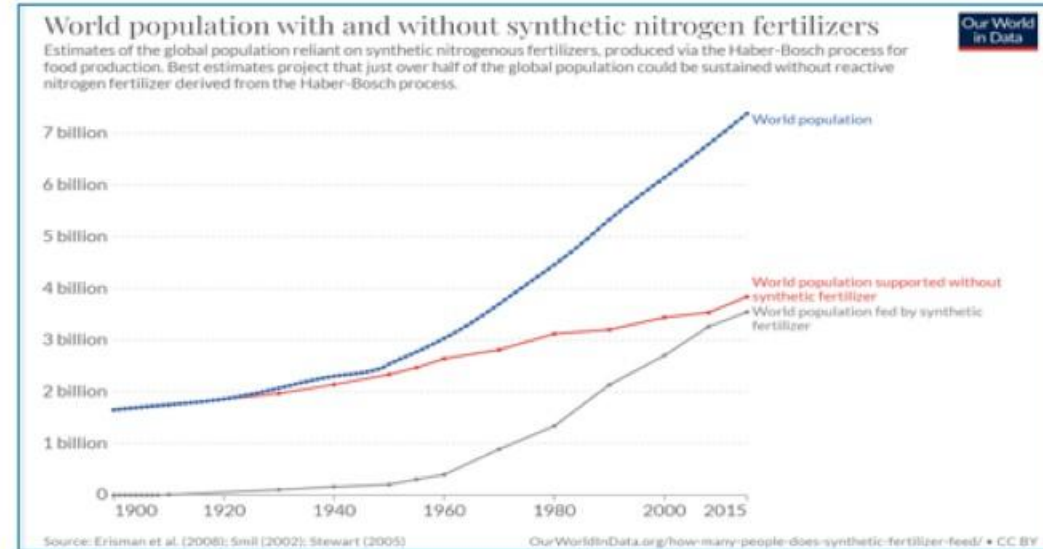
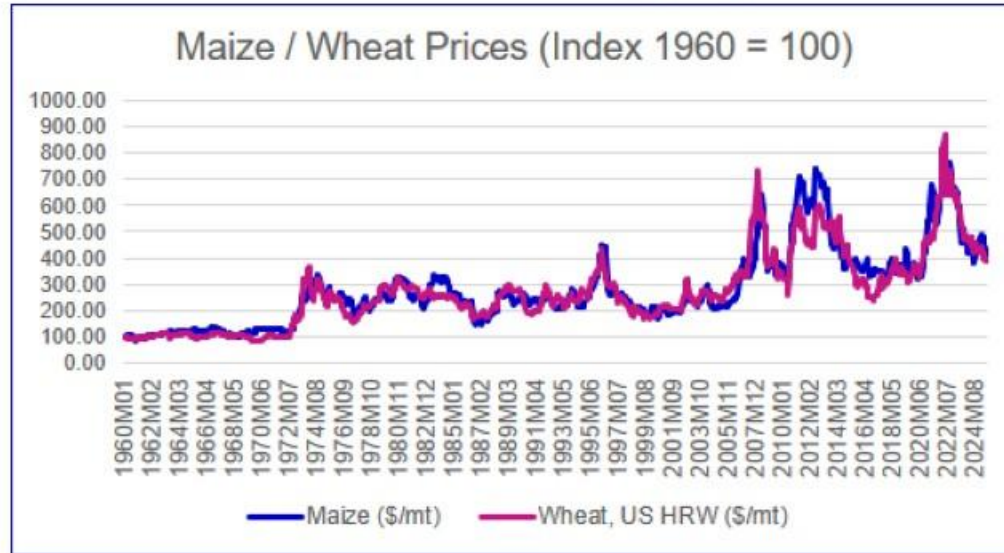
Increasing nitrogen fertilization could pollute rivers and worsen water scarcity



02/06/2024 – In the 2050s, one-third of the world's rivers could be affected by water scarcity or be polluted by nitrogen. This is the result of a study by an international team of scientists, including researchers from the Potsdam Institute for Climate Impact Research and led by scientists of Wageningen University. They assessed the scarcity of clean water under ongoing climate change and show how water pollution from intensified agriculture and untreated sewage could limit clean water supply.

Nitrogen pollution from agriculture could lead to water scarcity. This applies first and foremost to parts of Asia, North America and Africa, but also to Europe.

Disruption in food and energy systems



Oil Price - from to \$18 / bbl to \$147 / bbl to \$68 / bbl

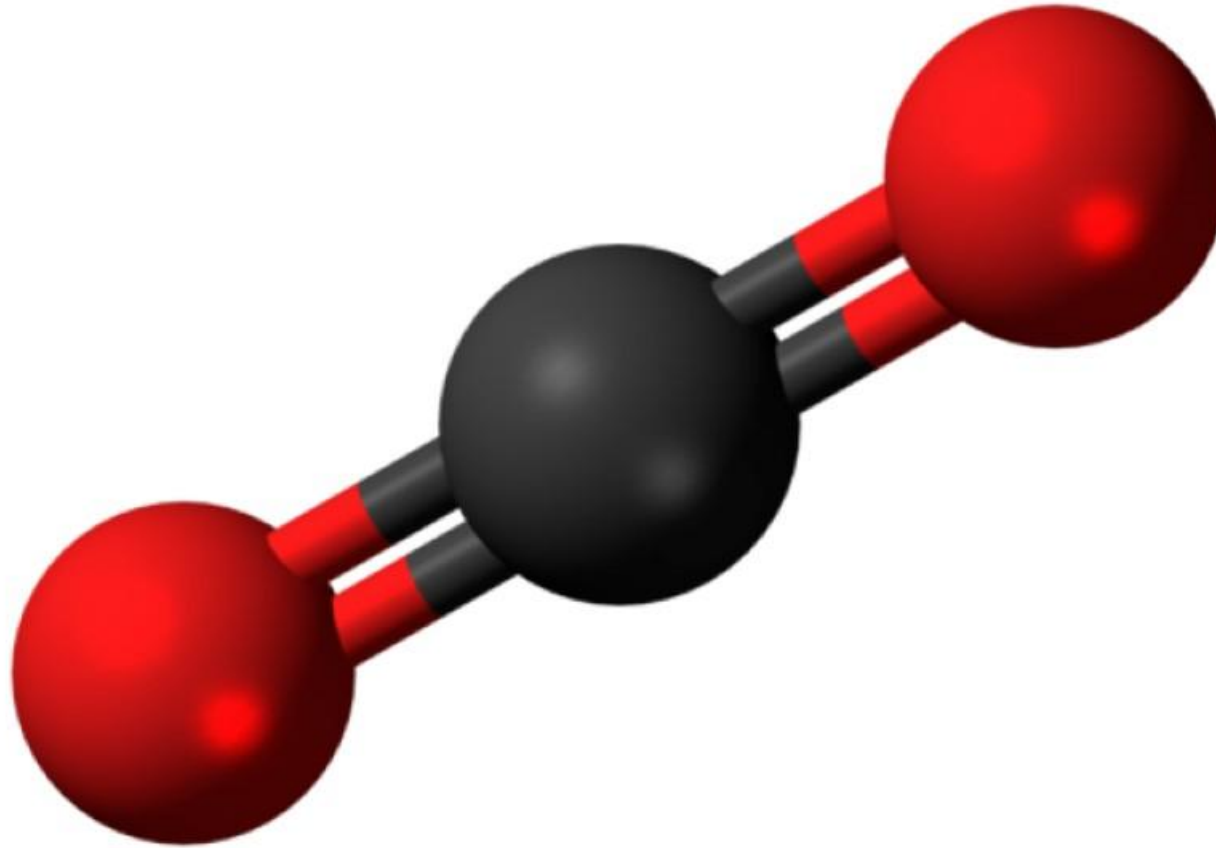
Europe Brent Spot Price FOB

Dollars per Barrel

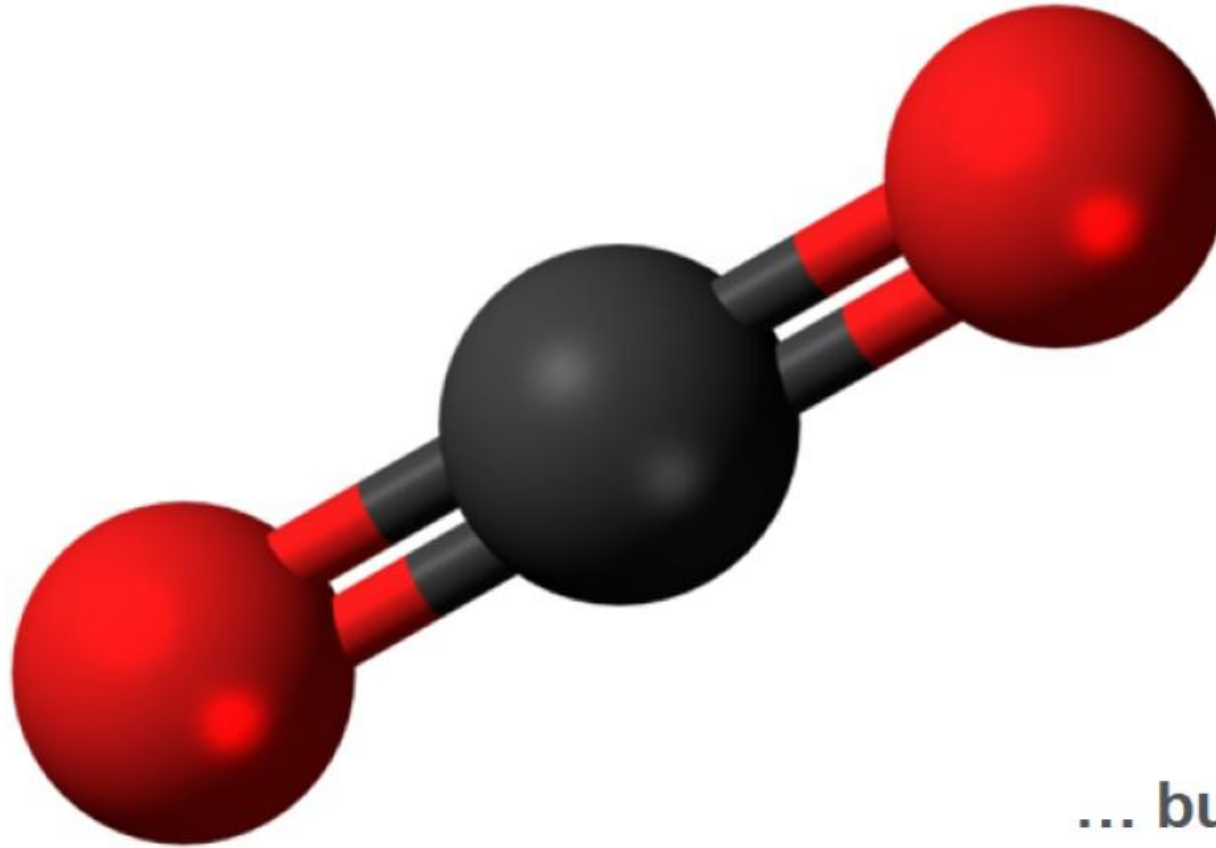


<https://www.eia.gov/dnav/pet/hist/LeafHandler.ashx?n=pets&sr=brte&f=m>

Enter carbon dioxide ... CO₂



Enter carbon dioxide ... CO₂



... but not quite yet

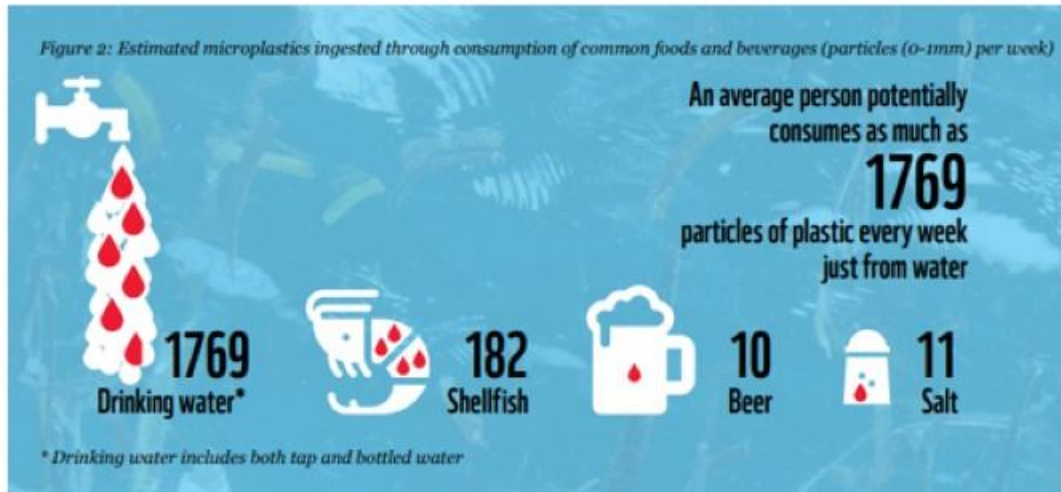
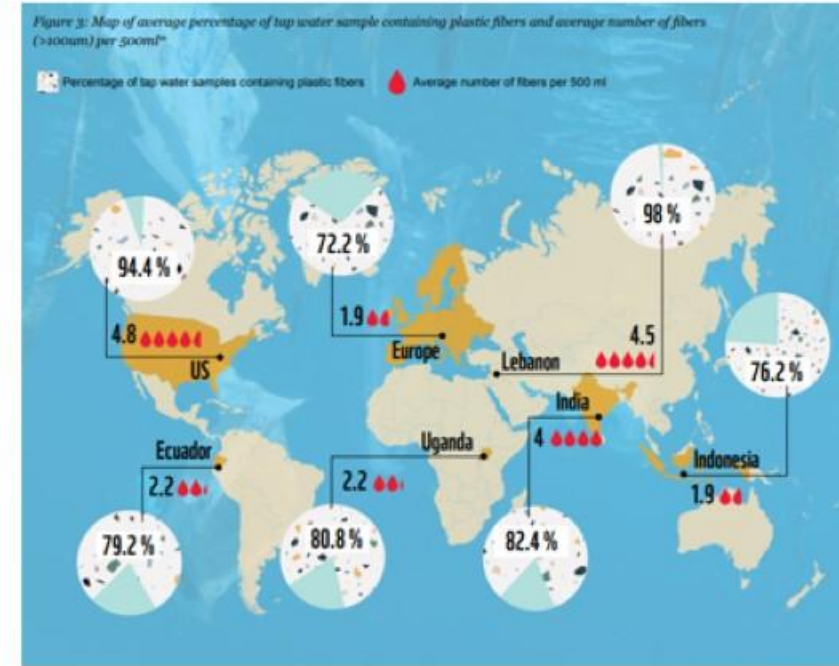
Sources and Sinks



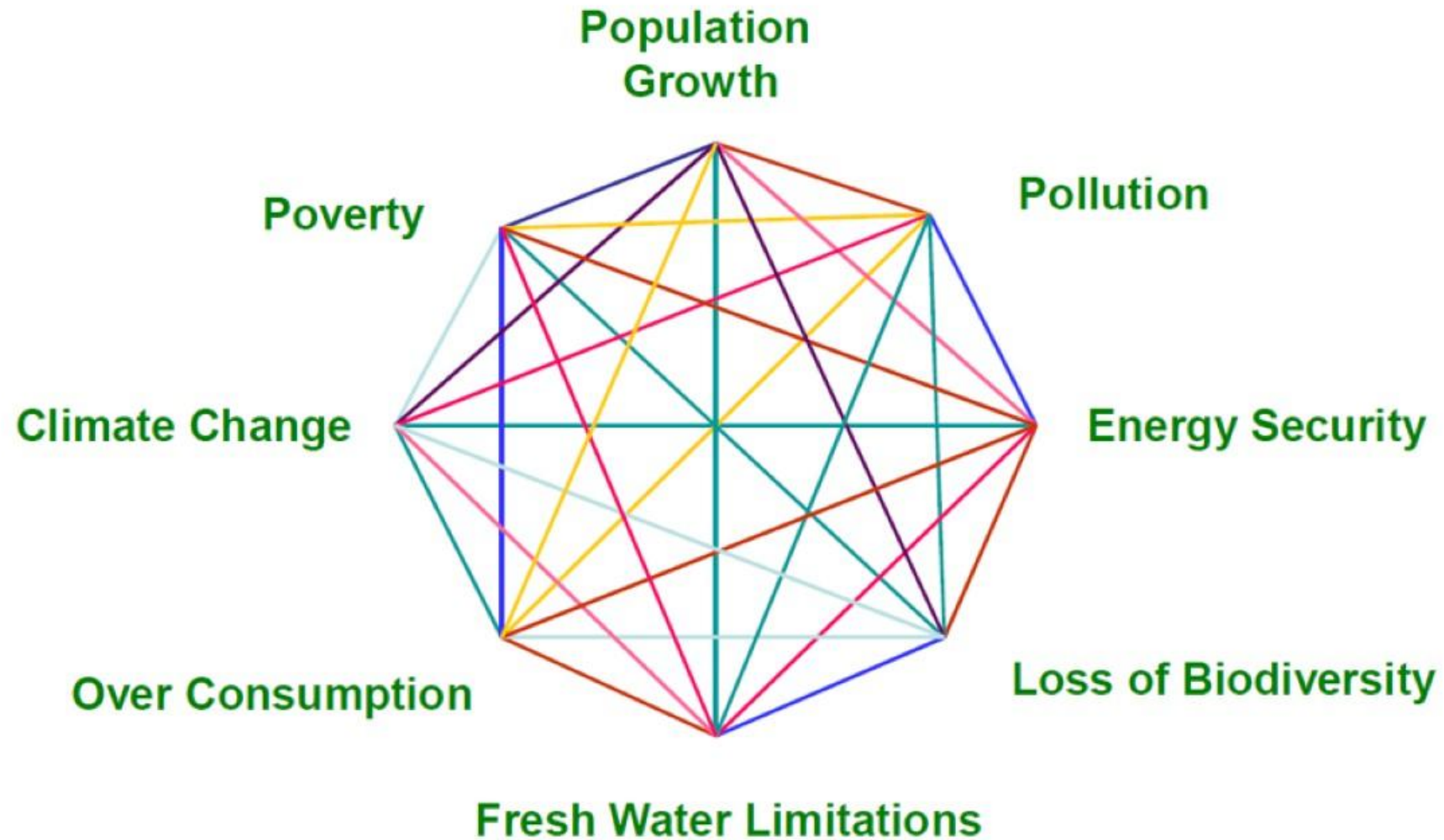
Sources and Sinks



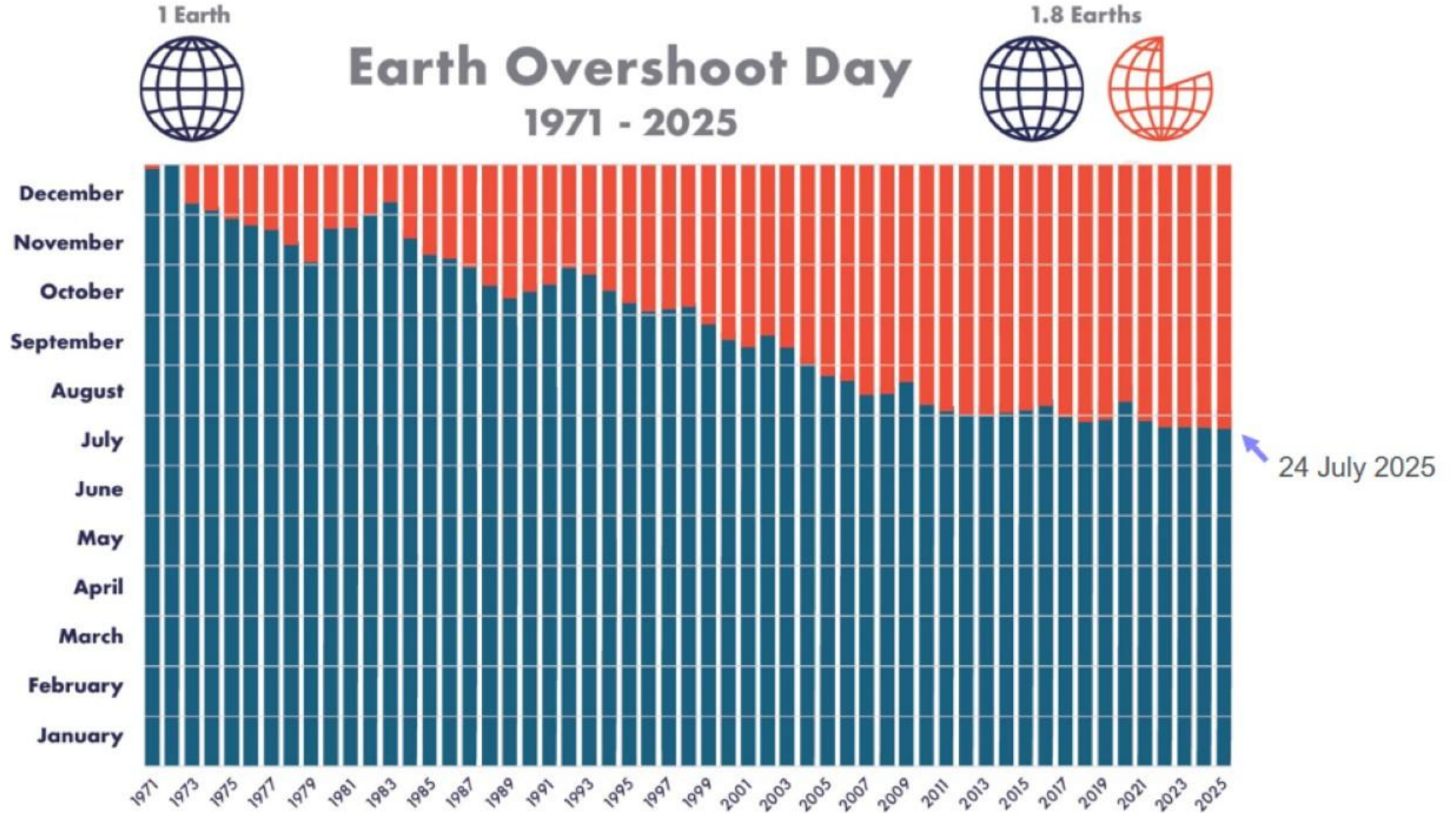
You eat a credit card's worth of plastic a week, research says



A Complex Interconnect Challenge



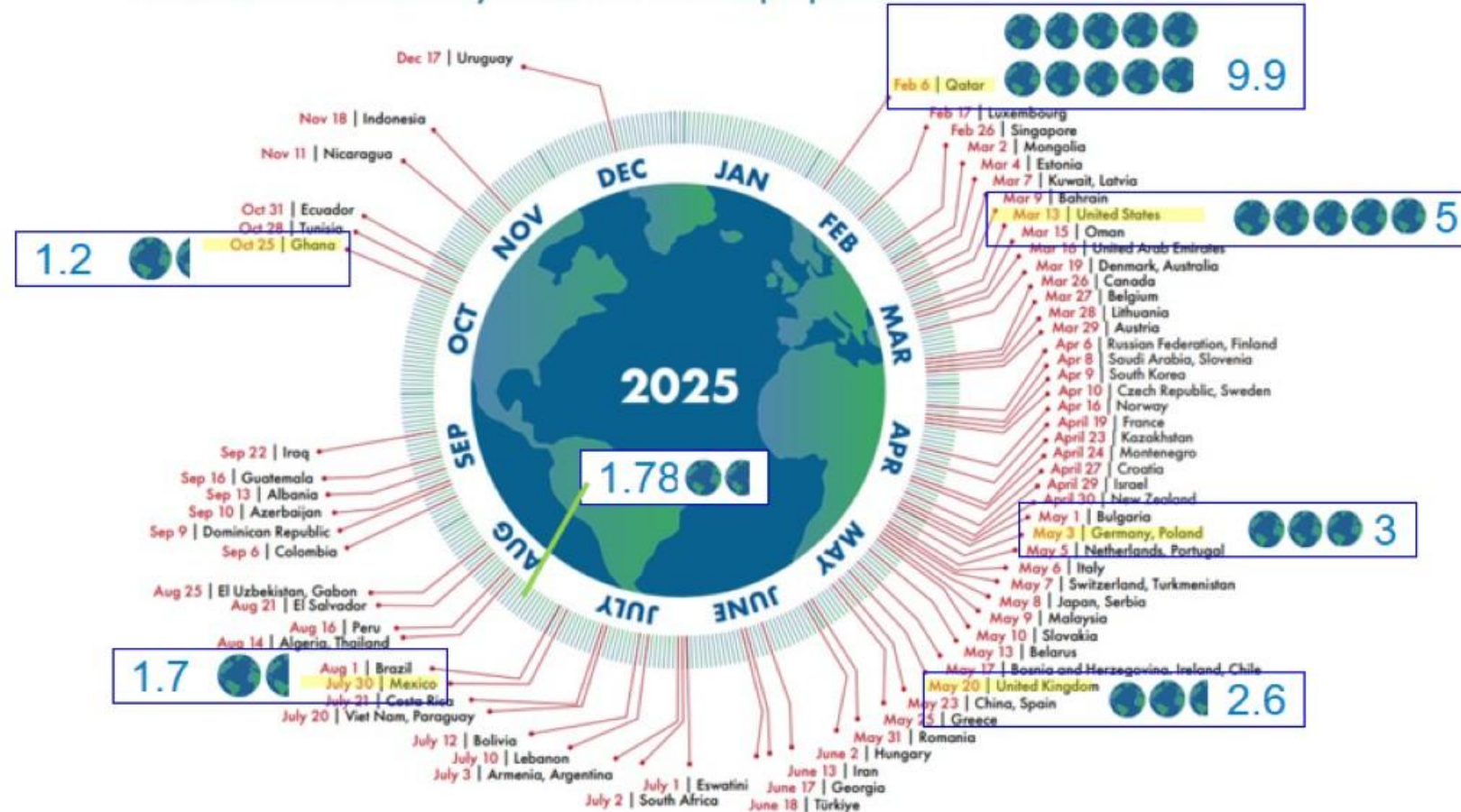
Earth overshoot day



Earth overshoot day

Country Overshoot Days 2025

When Earth Overshoot Day would land if all the people around the world lived like...



For more information, visit:
<https://overshootday.org/newsroom/country-overshoot-days/>

Source: National Footprint and Biocapacity Accounts, preliminary 2025 Edition
 York University, FoDaFo, Global Footprint Network, data.footprintnetwork.org



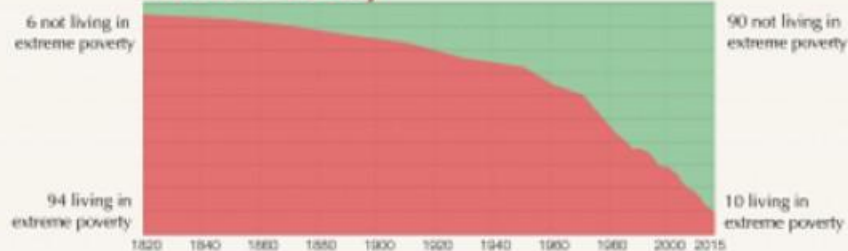
Source: <https://overshoot.footprintnetwork.org/newsroom/country-overshoot-days/>



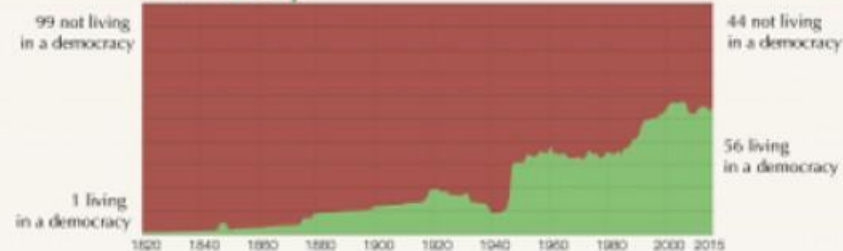
The World as 100 People over the last two centuries

Our World
in Data

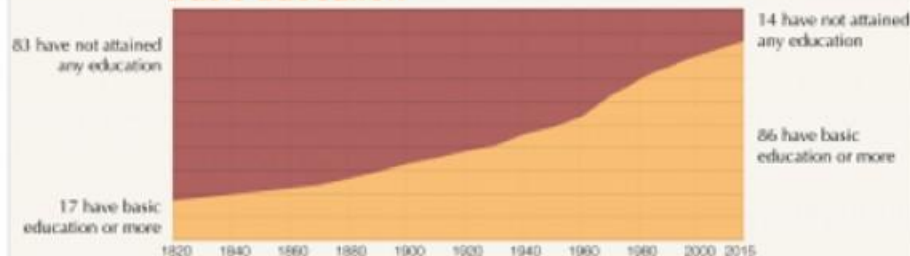
Extreme Poverty



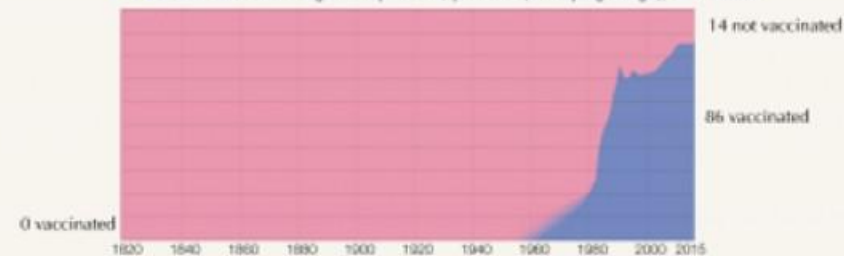
Democracy



Basic Education



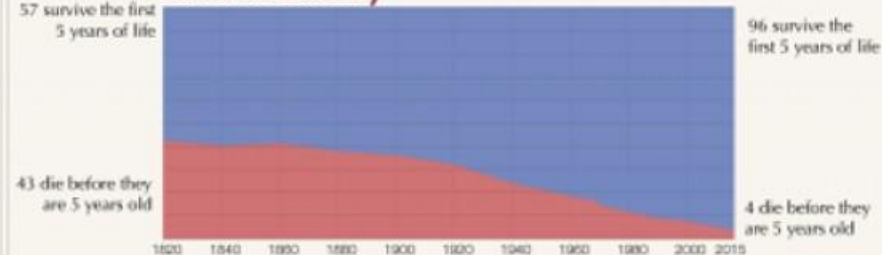
Vaccination against diphtheria, pertussis (whooping cough), and tetanus



Literacy



Child Mortality



Data sources:

Extreme Poverty: Bourguignon & Morrisson (2002) up to 1970 – World Bank 1981 and later (2015 is a projection).
Vaccination: WHO (Digital data are available for 1980 to 2015 – the DPT3 vaccination was licensed in 1949).
Education: OECD for the period 1820 to 1960, IEA for the time thereafter.
Literacy: OECD for the period 1820 to 1990, UNESCO for 2004 and later.

Democracy: Polity IV index (own calculation of global population share).
Colonisation: Blomster and Min (own calculation of global population share).
Continuity: HYDE database.
Child mortality: up to 1980 own calculations based on Geopendix; World Bank thereafter.



All these visualizations are from [OurWorldInData.org](https://ourworldindata.org) an online publication that presents the empirical evidence on how the world is changing.

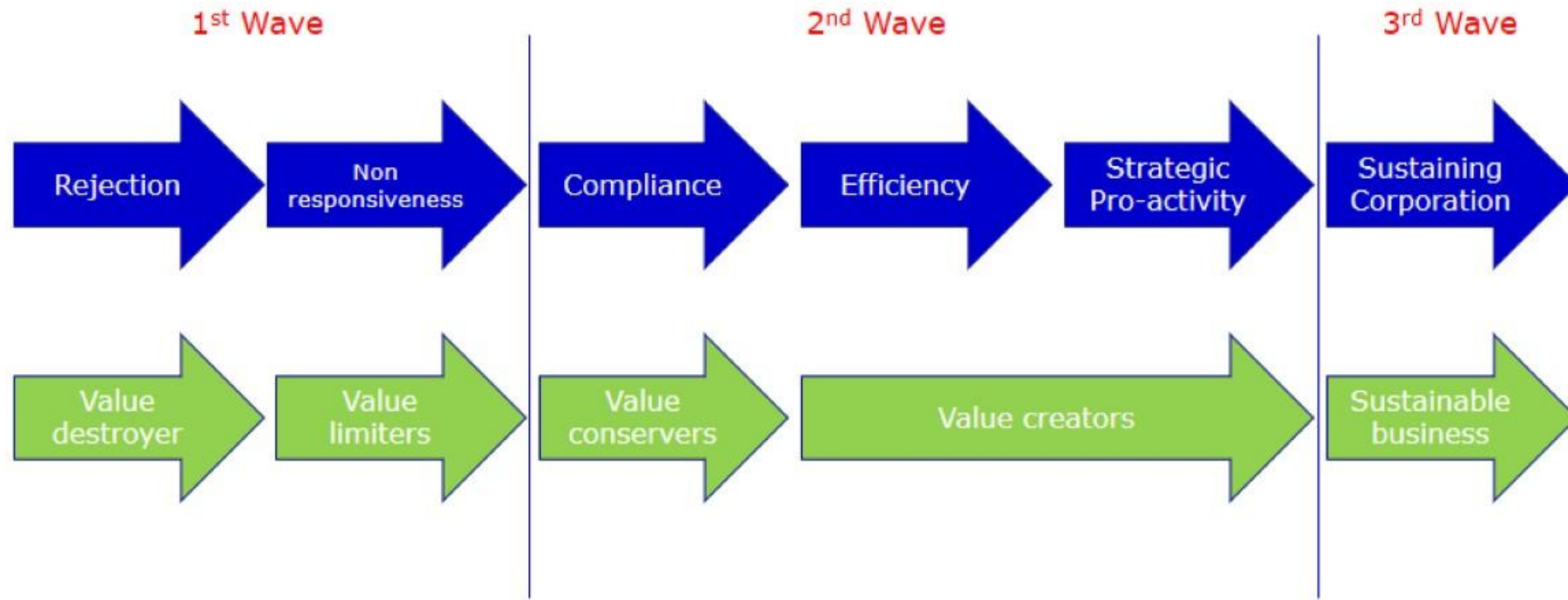
Licensed under CC-BY-SA by the author Max Roser

Sustainability

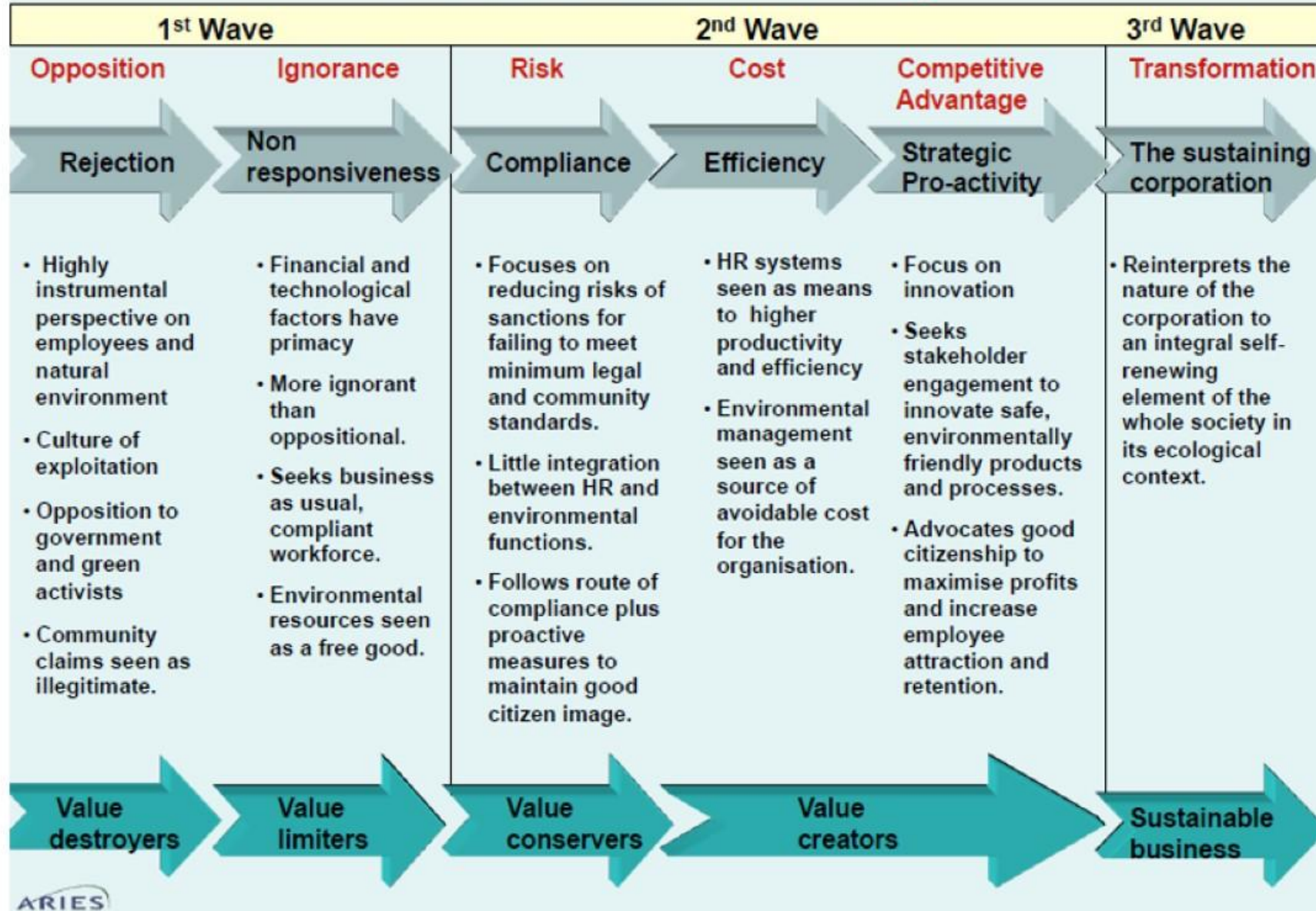
An Organisational Context

The Organizational Response

(Dunphy et al., 2003)



Waves of Sustainability



Exercise

Using Dunphy et al.'s (2003) model of sustainability:

- Identify 2 companies you consider to be in each of the three phases

Stay in touch...



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See you next time

Stay in touch:

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